Practice workbook · Greatest or least value in a region

Simple linear programming: finding the greatest or least value of an expression such as P = 2x + y over the region R.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-07

1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

5 (a) On the grid opposite draw the line with equation

(i) $y = 2x + 5$ (ii) $4y = x - 8$ (iii) $5y + 3x = 30$

(3)

(b) Show, by shading, the region R defined by the inequalities

$y \leqslant 2x + 5$ $4y \geqslant x - 8$ $5y + 3x \leqslant 30$

(1)

For all points in R with coordinates (x, y)

$P = 2x - 5y$

(c) Using your graph, find the least value of P

(3)

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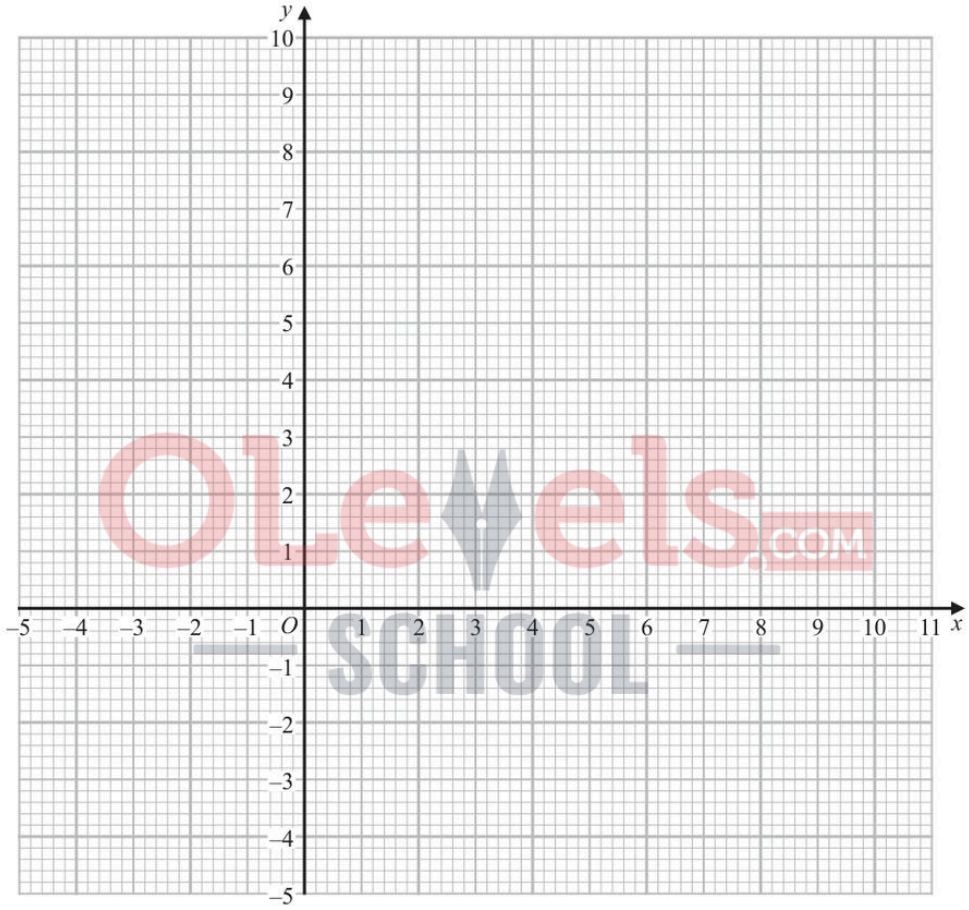
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1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued



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Turn over for a spare grid if you need to redraw your graph.

1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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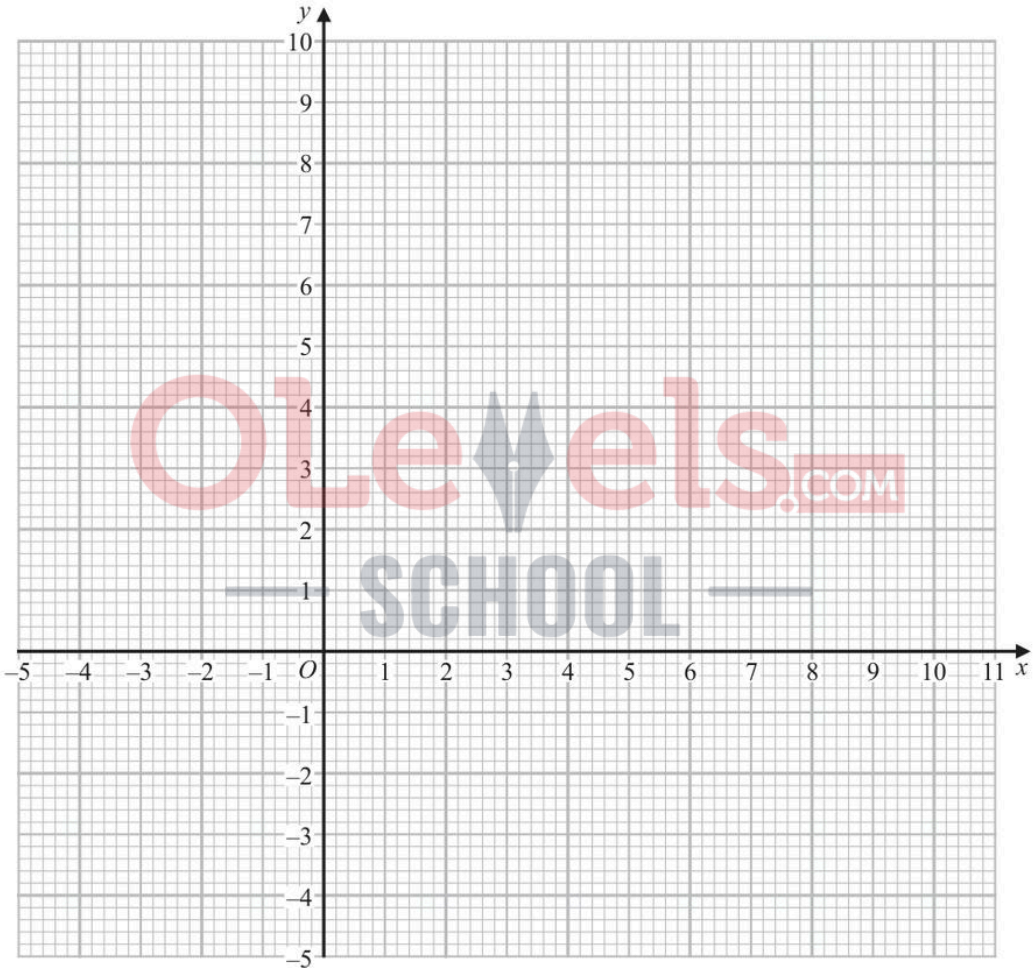
1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 7 marks)

1. Question 5 · part c · 3 marks · Working and answer

Working and answer

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2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

4 (a) On the grid opposite, draw the line with equation

(i) $y = 2x - 4$ (ii) $2x + 3y = 12$ (iii) $y + 2x + 2 = 0$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$y \geqslant 2x - 4$ $2x + 3y \leqslant 12$ $y + 2x + 2 \geqslant 0$

(1)

For all points in R, with coordinates (x, y)

$$P = x - 2y$$

(c) find the least value of P

(4)

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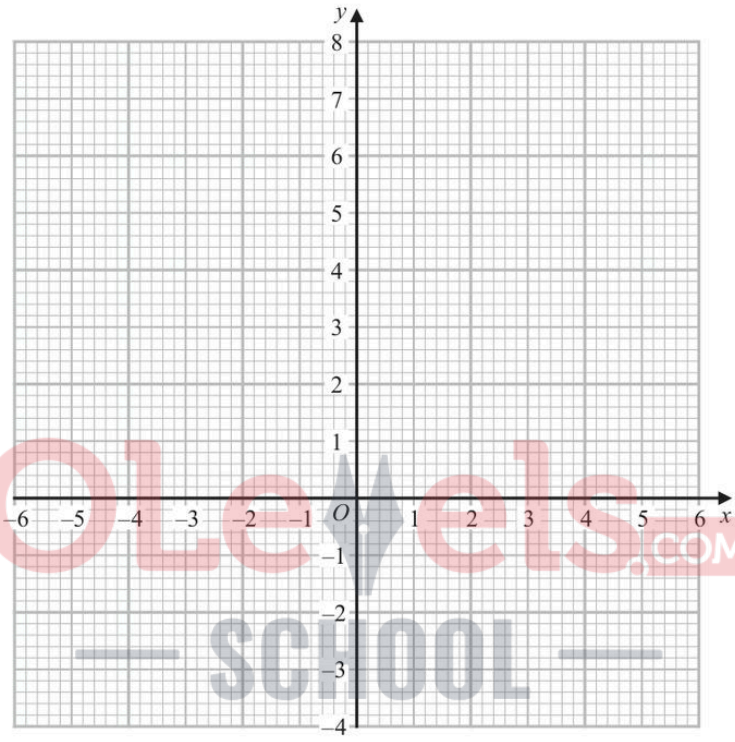
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2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 4 continued



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P 6 6 3 1 0 A 0 9 3 6

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Turn over ▶

2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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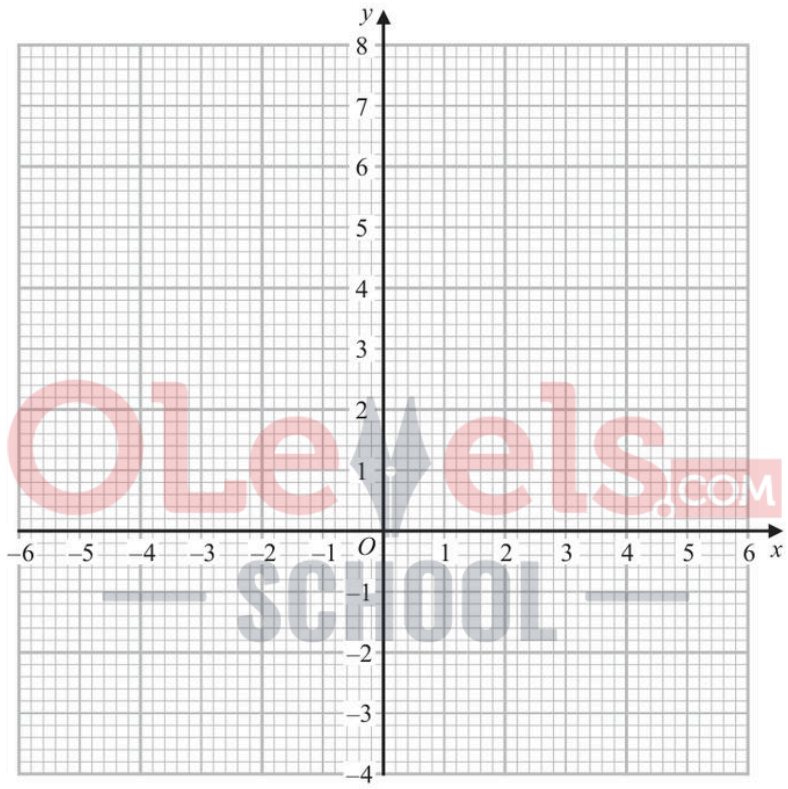
2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 8 marks)

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2. Question 4 · part c · 4 marks · Working and answer

Working and answer

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3. Question 4 · part c.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part c.ii only.

4 (a) On the axes opposite, draw the line with equation

(i) $y = -x - 1$ (ii) $y - 3x + 8 = 0$ (iii) $2y = x + 8$

(3)

(b) Show, by shading on your graph, the region R defined by the inequalities

$y \geqslant -x - 1$ and $y \geqslant 3x - 8$ and $2y \leqslant x + 8$

(1)

For all points in R, with coordinates (x, y)

$$P = 2y - 3x$$

(c) Find

(i) the greatest value of P

(ii) the least value of P

(4)

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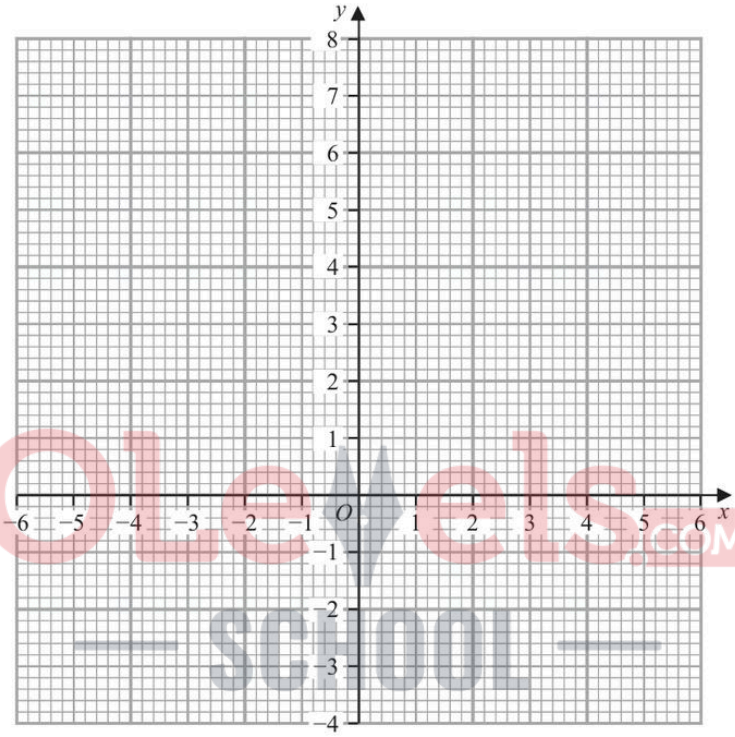
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3. Question 4 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.

Question 4 continued



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3. Question 4 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.

Question 4 continued

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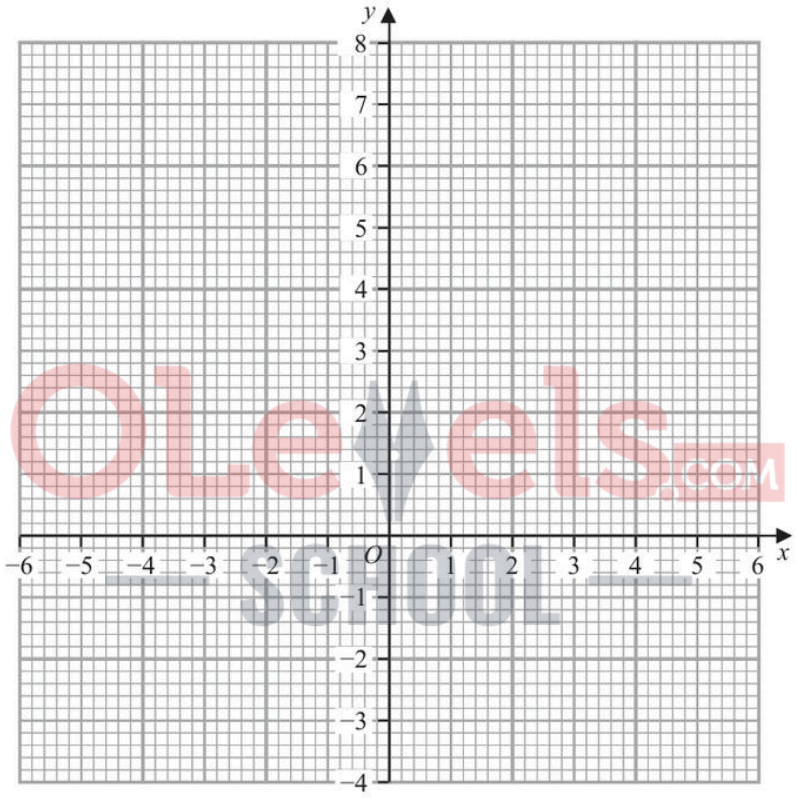
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Further practice · Attempt part c.ii only.

Question 4 continued

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(Total for Question 4 is 8 marks)

3. Question 4 · part c.ii · 4 marks · Working and answer

Working and answer

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4. Question 4 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

4 (a) On the grid opposite, draw the graph of the line with equation

(i) $3x+2y=18$ (ii) $x+3y-6=0$ (iii) $y=3x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x+2y \leqslant 18$ $x+3y-6 \geqslant 0$ $y \leqslant 3x$

(1)

For all points in R, with coordinates (x, y)

$$P=2x-y$$

(c) find the least value of P

(4)

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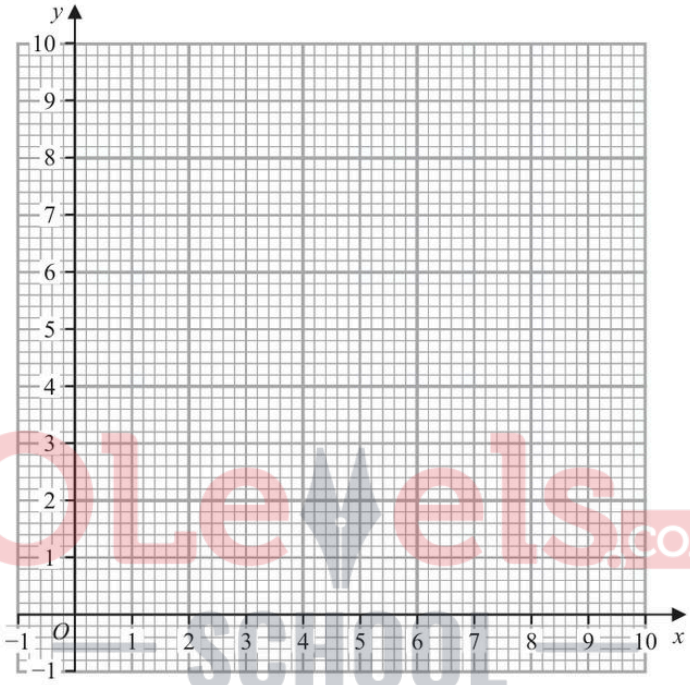
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4. Question 4 · part c · 4 marks

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Further practice · Attempt part c only.

Question 4 continued



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4. Question 4 · part c · 4 marks

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Question 4 continued

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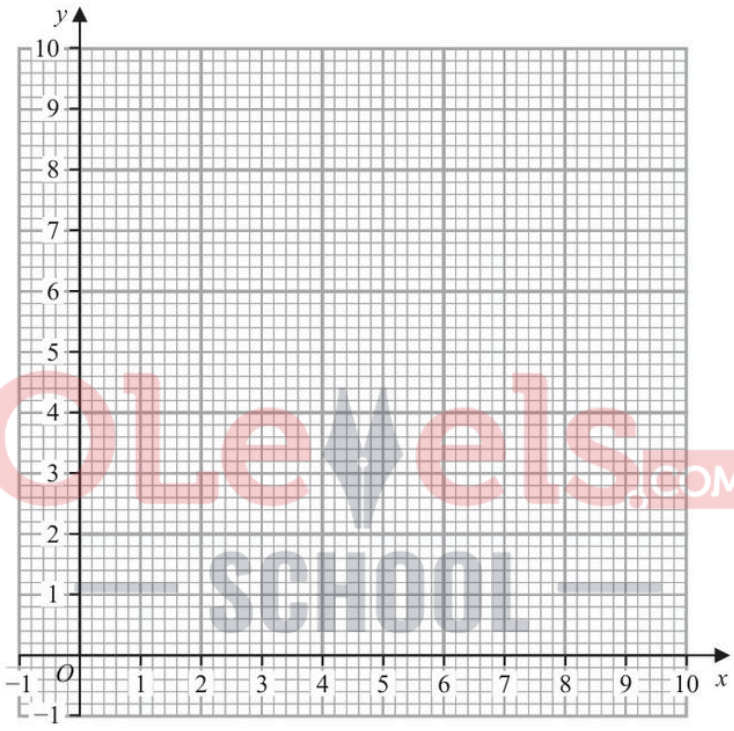
4. Question 4 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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4. Question 4 · part c · 4 marks · Working and answer

Working and answer

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5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 (a) On the grid opposite, draw the graphs of the lines with equations

$$2x + 3y = 24 \qquad y = 2x \qquad 3y = 2x - 12$$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$$2x + 3y \leqslant 24 \qquad y \leqslant 2x \qquad 3y \geqslant 2x - 12 \qquad y \geqslant 0$$

(1)

For all points in R, with coordinates (x, y)

$$F = 2x + 5y$$

(c) Find the greatest value of F.

(3)

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5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

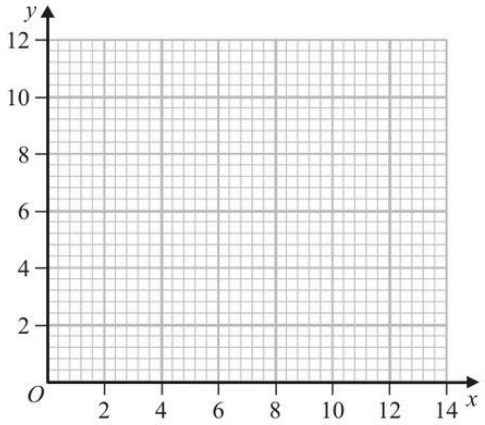
Further practice · Attempt part c only.

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Question 5 continued



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5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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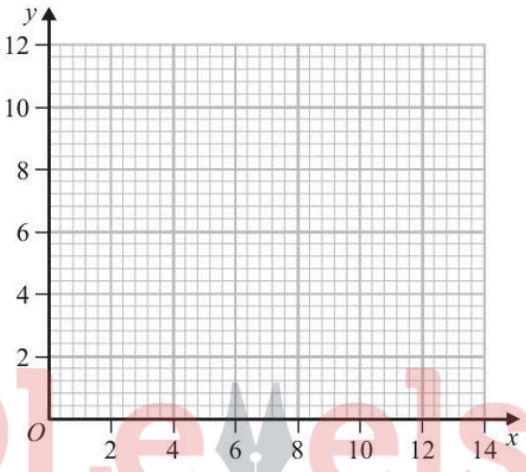
5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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5. Question 5 · part c · 3 marks · Working and answer

Working and answer

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6. Question 5 · part c · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

5 (a) On the grid opposite, draw the line with equation

(i) $3x-4y=12$ (ii) $y+6+3x=0$ (iii) $3y=18-x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x-4y \leqslant 12$ $y+6+3x \geqslant 0$ $3y \leqslant 18-x$

(1)

For all points in R, with coordinates (x, y)

$$P=3x-2y$$

Using values from your graph,

(c) find the least value of P and the greatest value of P

(4)

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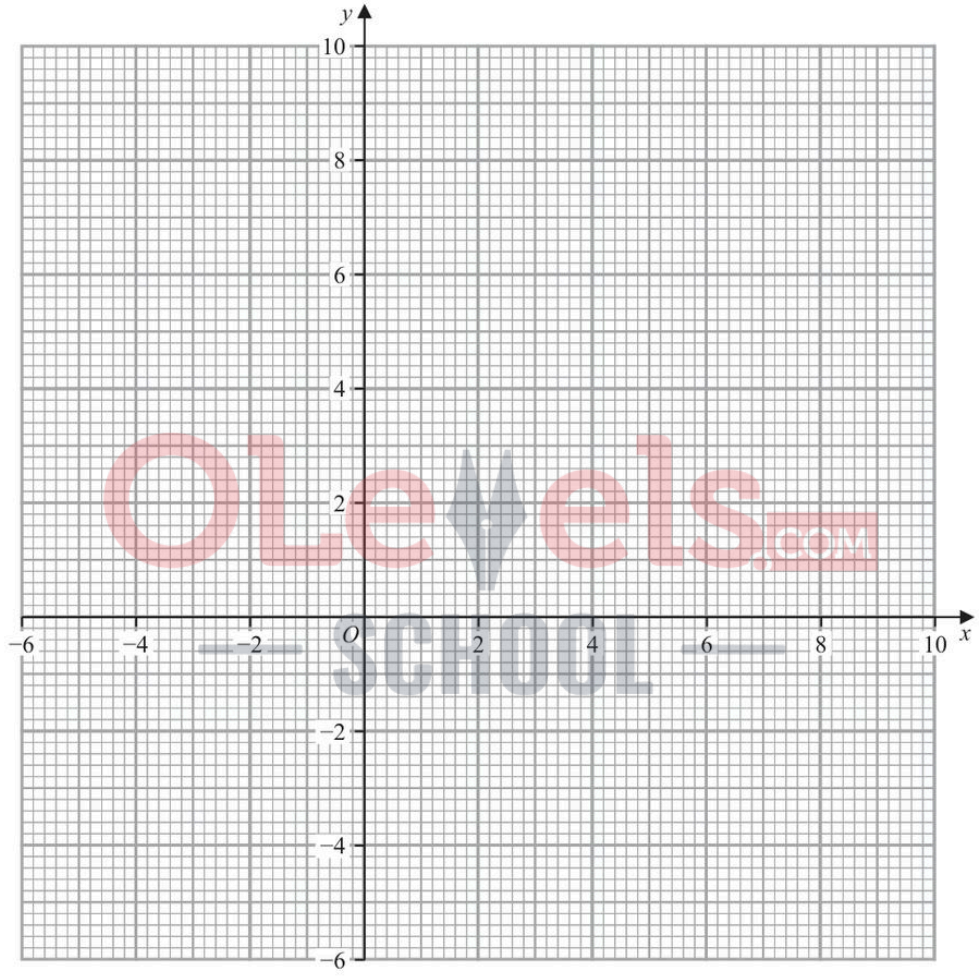
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6. Question 5 · part c · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

Question 5 continued



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6. Question 5 · part c · 4 marks

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Further practice · Attempt part c only.

Question 5 continued

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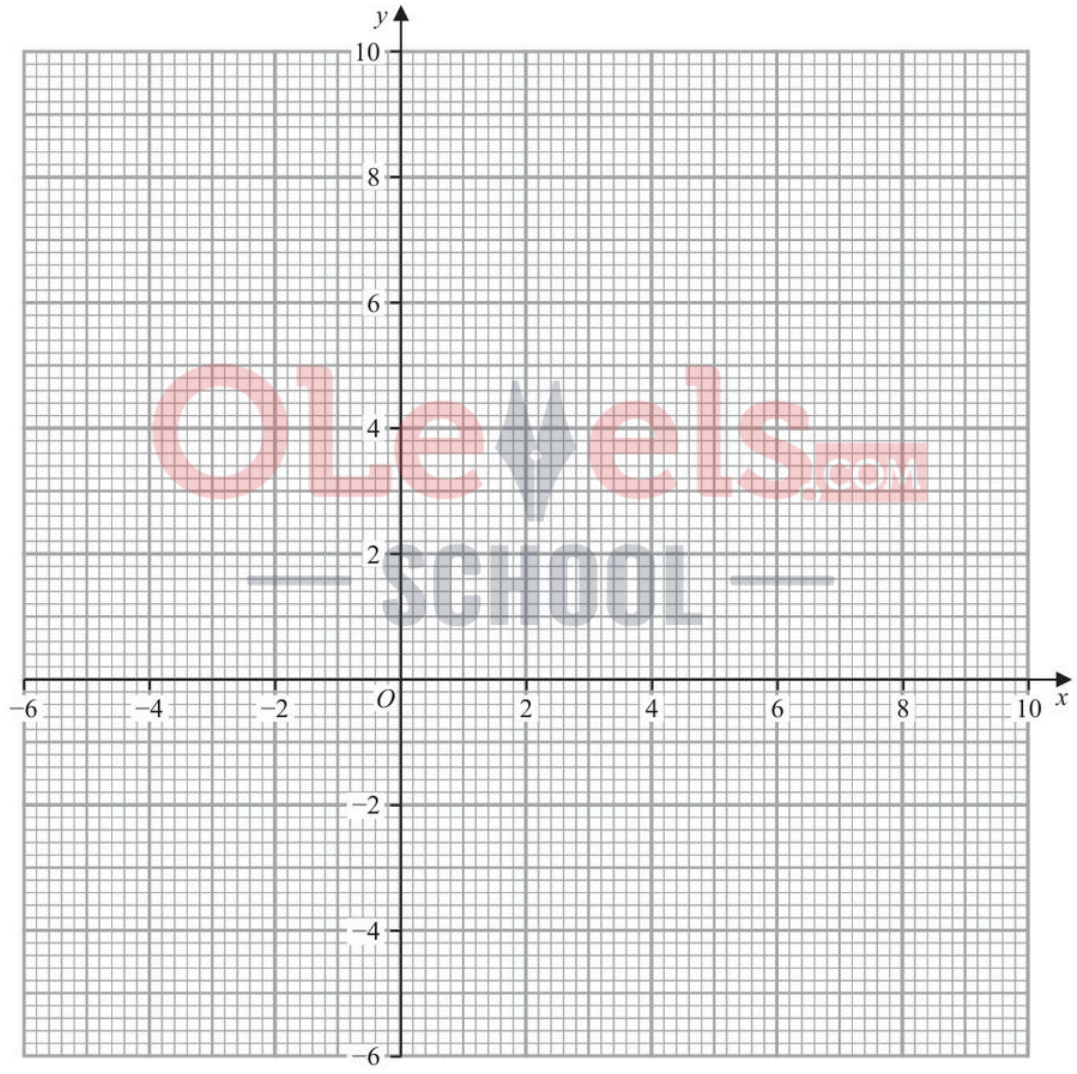
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Question 5 continued

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6. Question 5 · part c · 4 marks · Working and answer

Working and answer

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